Checking understanding of arithmetic procedures with integers and decimals

Task A: Arithmetic procedures with integers

1) Find the missing digits in each of the following:

a)
$$\begin{array}{r} \boxed{1}4\,6 \\ +3\,9\boxed{4} \\ \hline 5\boxed{4}0 \\ \hline \end{array}$$

b)
$$\begin{array}{r} 7\,3\boxed{5} \\ -3\boxed{8}7 \\ \hline \boxed{3}4\,8 \\ \hline \end{array}$$

c)
$$\begin{array}{r} 4\boxed{6}2\,5 \\ +\boxed{6}0\boxed{9}2 \\ \hline \boxed{1}0\,7\,1\boxed{7} \\ \hline \end{array}$$

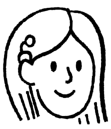
d)
$$\begin{array}{r} \boxed{6}4\,1\boxed{8} \\ -4\boxed{3}2\,5 \\ \hline 2\,0\boxed{9}3 \\ \hline \end{array}$$

2) Calculate the following, without a calculator. Try to use the efficient methods we have talked about.

a) 580×5 $= 580 \times 10 \div 2 = 5800 \div 2 = 2900$
b) 87×4 $= 87 \times 2 \times 2 = 174 \times 2 = 348$
c) 26×8 $= 26 \times 2 \times 2 \times 2 = 52 \times 2 \times 2 = 104 \times 2 = 208$
d) 45×6 $= 45 \times 3 \times 2 = 135 \times 2 = 270$
e) $485 \div 5$ $= 485 \div 10 \times 2 = 48.5 \times 2 = 97$

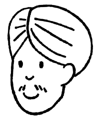
Task B: Addition and subtraction of decimals

1) Sofia and Andeep record how far they kick a football.
a) What is the total distance? 22.53 metres
b) How much further did Sofia kick the ball? 2.93 metres

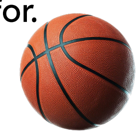


I kicked it 12.73 metres.

I kicked it 9.8 metres.



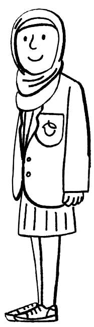
2) Lucas, Aisha and Jun record how long they can bounce a basketball for.



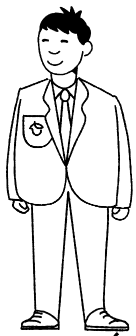
a) What is the total time? 91.92 seconds
b) How much longer did Lucas manage than Jun? 7.73 seconds

I did 35.6 seconds.

I did 28.45 seconds.



And, I managed 27.87 seconds.



3) Here are three slightly different subtractions.

a)
$$\begin{array}{r} 37.4 \\ -\ 2\boxed{7}.3 \\ \hline \boxed{1}0.\boxed{1} \\ \hline \end{array}$$

b)
$$\begin{array}{r} 37.4 \\ -\ 2\boxed{6}.5 \\ \hline \boxed{1}0.\boxed{9} \\ \hline \end{array}$$

c)
$$\begin{array}{r} 37.4 \\ -\ 2\boxed{8}.5 \\ \hline \boxed{0}8.\boxed{9} \\ \hline \end{array}$$

Without calculating, explain why the following statements are false.

a) The digit in the ▯ must be 1 in each calculation as the tenths digits are one apart.
True for a but not b & c, as we subtract the bottom digit from the top

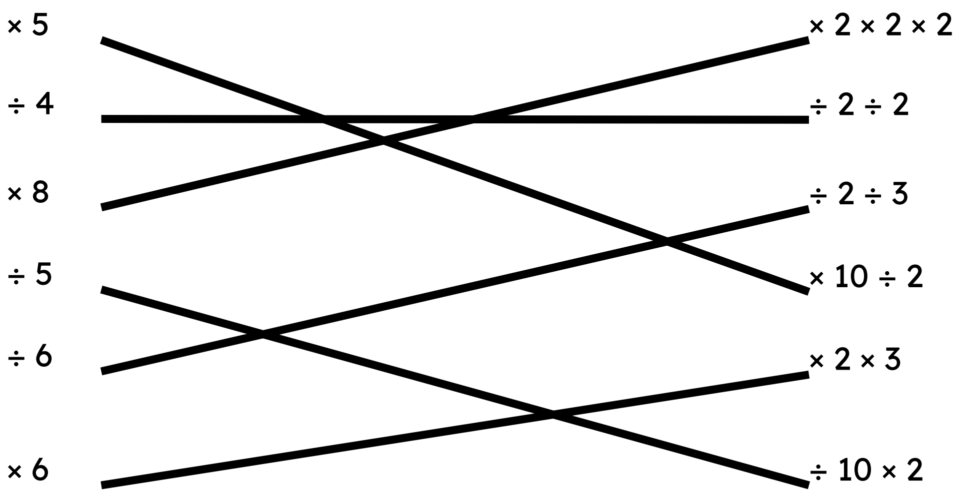
b) The digit in the △ must be 7 in both cases as the difference is 0. *True for a but not b, as the 1 of the ones would have been exchanged for 10 tenths*

c) The digit in the ☆ must be 1 in all three calculations as 3 – 2 = 1. *True for a & b but not c, as 1 of the tens would have been exchanged for 10 ones*

d) Find the missing digits.

Task C: Multiplication and division of decimals

1) Match the operation on the left to its equivalent on the right.



2) Find the cost of the following:

a) 2 boxes *£9.50*
b) 10 boxes *£47.50*
c) 5 boxes *£23.75*
d) 6 boxes *£28.50*

3) The Oak students decide to buy one box of chocolates between them.

Work out how much each person pays if:

a) 2 students buy the box *£2.90*
b) 4 students buy the box *£1.45*
c) 5 students buy the box *£1.16*